Short answers test

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Short Answers

1) **Sampling :-**

- Low spatial resolution occurs when the image is sampled with fewer pixels. Small objects lose important details and appear unclear.
- Increase the sampling rate or use a higher resolution camera, this provides more pixels and improves visibility of small objects.

2). **Dark and noisy Image :-**

- Use histogram equalization to improve the brightness and contrast.
- A median filter can be used to reduce noise in the image. These techniques enhance important details in dark areas.

3). **Quantization :-**

- Quantization converts continuous intensity values into discrete values.

For example, intensity values can be represented using levels from 0 to 255. It reduces the infinite range of intensity values to finite digital levels. Aws, Image can store and process by computer.

4). **Uneven Lighting:-**

- Use contrast enhancement to improve visibility of product defects. Adaptive histogram equilization can enhance contrast in different image regions.

Therefore, the inspection system can identify defects more easily.

5). **Image Segmentation:-**

- Image segmentation divides an image into meaningful regions. The road vehicles, pedestrain and background can be seperated.

It supports obstacle detection and safe autonomous navigation.

6). Edge detection :- By applying the edge detection technique such as canny edge detector.

- It can identifies sudden changes in intensity representing the object edges and boundaries. Cracks appear as edges and can identified.

7). Filtering :- A median filter can be used to remove unwanted noise from the medical image. It replaces noisy pixels using the median value of neighbouring pixels. It also reduces noise while preserving important edges.

- Therefore, the tumor becomes clear and easier to identify.

8). Object detection :-

Object detection identifies vehicles present in the traffic-monitoring image. Each vehicle can be located using bounding bones and identified.

- This helps in traffic counting, congestion monitoring and management. The vehicles can be tracked across video frames.

9). Motion blur :-

Motion blur is an main cause of movement of object or camera during acquistion. This produces motion blur and it reduces the image sharpness. Using the faster shutter speed can also be reduce the motion blur.

10). Image classification :- Image classification can classify the manfactured products as good or defective. The system analyze features such as shape, texture and appearence. This enables the fast and consistent automated quality inspection.